
ECON 425 - ADVANCED ECONOMETRICS

计量经济学进阶

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1 Review on Probability

For our study in econometrics, it is important to lay down the groundwork in probability theory. Therefore, we start by giving a general review on probability theory, and highlight the concepts where we use most frequently in the field of econometrics.

1.1 Basic Concepts

Definition 1.1. Let S be a sample space which denotes a set of all possible outcomes.

where each element $s \in S$ is an outcome. Then, we consider a σ -field $\mathcal{F}$ which is a collection of subsets of S , such that it satisfies these properties:

1. $S \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then the complement of A in S is also an element of $\mathcal{F}$.
3. If $A_1, A_2, \dots, A_n \in \mathcal{F}$, then $\bigcup_{i=1}^n A_i \in \mathcal{F}$.

With this set, consider a probability map $\mathbb{P}$ such that

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$$

Definition 1.2. $(S, \mathcal{F}, \mathbb{P})$ forms a probability space.

Definition 1.3. A random variable X is a map

$$X : S \rightarrow \mathbb{R}$$

Analogously, a random vector is a map $X : S \rightarrow \mathbb{R}^k$.

Consider a set B , we write:

$$\mathbb{P}(\{s \in S \mid X(s) \in B\}) = \mathbb{P}(X \in B)$$

A map that maps B to $\mathbb{P}(X \in B)$ describes the distribution of X .

Proposition 1.1. If random variables/vectors are identically distributed, $f(X_i)$ and $f(X_j)$ are also identically distributed for some transformation f .

Definition 1.4. Consider two random variables/vectors X_1, X_2 , and given two sets A, B . If

$$\mathbb{P}(X_1 \in A \mid X_2 \in B) = \mathbb{P}(X_1 \in A)$$

or

$$\mathbb{P}(X_1 \in A \wedge X_2 \in B) = \mathbb{P}(X_1 \in A) \times \mathbb{P}(X_2 \in B)$$

we say the two random variables/vectors X_1, X_2 are independent.

Proposition 1.2. If X_i and X_j are independent, then $f(X_i)$ and $g(X_j)$ are independent for transformations f, g and random variables/vectors X_i, X_j .
 Additionally, $\text{Cov}(X_i, X_j) = 0$.

However, note that, two **uncorrelated** random variables/vectors does not imply they are independent. We give some useful properties about variance and covariance:

1. $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$.
2. $\text{Cov}(aX + bY, cZ + dW) = ac\text{Cov}(X, Z) + ad\text{Cov}(X, W) + bc\text{Cov}(Y, Z) + bd\text{Cov}(Y, W)$.
3. $\text{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y))) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$.
4. $\text{Var}(X) = \text{Cov}(X, X)$.

Additionally, for independent identically distributed (abbr. iid) variables:

1. $\mathbb{E}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \mathbb{E}(X_i) = n\mathbb{E}(X_i)$.
2. $\text{Var}(\sum_{i=1}^n X_i) = n\text{Var}(X_i)$.

Now we discuss conditional expectation. Consider a continuous random variable X with probability distribution of f_X so that

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

Since we have the probability map of $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ and an event C , then we have the map of $\mathbb{P}_C : \mathcal{F} \rightarrow [0, 1]$ where

$$\mathbb{P}_C(A) = \mathbb{P}(A | C) = \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(C)}$$

Then, given some new information Y , the joint density is thus: $f_{X,Y}(x, y)$, with the conditional density of X given $Y = y$ to be:

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

This means that $\mathbb{E}(X | Y = y)$ is a **constant** and a function of y : $g_{X|Y}(y)$. We then write $\mathbb{E}(X | Y) = g_{X|Y}(Y)$, which is a **random variable**.

Several properties of the conditional expectation include:

1. $\mathbb{E}(\mathbb{E}(X | Y = y)) = \mathbb{E}(X | Y = y)$.
2. $\mathbb{E}(aX_1 + bX_2 | Y) = a\mathbb{E}(X_1 | Y) + b\mathbb{E}(X_2 | Y)$.
3. If X, Y are independent, we have $\mathbb{E}(X | Y) = \mathbb{E}(X)$.
4. $\mathbb{E}(Xg(Y) | Y) = g(Y)\mathbb{E}(X | Y)$.
5. The **Law of Iterated Expectation** states that $\mathbb{E}(\mathbb{E}(X | Y)) = \mathbb{E}(X)$. Or more generally, we have

$$\mathbb{E}(\mathbb{E}(X | \text{large info.}) | \text{small info.}) = \mathbb{E}(X | \text{small info.})$$

$$\mathbb{E}(\mathbb{E}(X | \text{small info.}) | \text{large info.}) = \mathbb{E}(X | \text{small info.})$$

where “small info.” means that you can obtain it from “large info.”. For example, knowing Z is a small info. compared to knowing X, Z .

This also gives conditional variance to be:

$$\text{Var}(Y \mid X) = \mathbb{E}(Y^2 \mid X) - \mathbb{E}(Y \mid X)^2$$

and conditional variance behaving in a similar way.

Lemma 1.1. For any transformation g , we have:

$$\mathbb{E}((Y - g(X))^2) \geq \mathbb{E}((Y - \mathbb{E}(Y \mid X))^2)$$

2 Linear Regression Model

A linear regression model is what we use to model correlation between economic objects of interest. Consider a univariate linear regression model to be:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

where we observe iid random vectors (X_i, Y_i) and we do not observe β_0, β_1, u_i . Given any model, we need to do the following things:

1. Clarify the object of interest.
2. Be clear about the meaning or interpretation of the object of interest.
3. Be clear about the assumptions we makes.

There are two kinds of uncertainty:

1. Identification failure, and
2. Estimation error.

2.1 Identification

To identify the parameters, we usually make three assumptions:

1. Exogeneity: $\text{Cov}(X_i, u_i) = 0$.
2. No multicollinearity: $\text{Var}(X_i) > 0$.
3. Normalization of the error: $\mathbb{E}(u_i) = 0$.

We can identify β_1 under the exogeneity assumption and the no multicollinearity assumption, and get that:

$$\beta_1 = \frac{\text{Cov}(Y_i, X_i)}{\text{Var}(X_i)}$$

Furthermore, we can identify β_0 under the normalization of error assumption where:

$$\beta_0 = \mathbb{E}(Y_i) - \beta_1 \mathbb{E}(X_i)$$

after identifying β_1 .

There are situations where these assumptions may fail, and different kinds of failures may lead to failure of Identification. However, in certain cases, even if certain assumptions fail, it is still possible to identify some parameters. The process of analysis remains the same, that is, figuring out how $\text{Cov}(X_i, u_i)$ acts. Additionally, if we were to show there is a failure of identification, we need to propose an alternative model and demonstrate that altering the parameters will still satisfy the conditions that we already observe.

The above assumptions discussed is used to identify the parameters of a univariate linear regression model. For multivariate linear regression model

$$Y_i = \beta_0 + \sum_{j=1}^n \beta_j X_{ij} + u_i$$

the identification assumptions are updated to be:

1. Exogeneity: $\forall X_{ij}, \text{Cov}(X_{ij}, u_i) = 0$.
2. No multicollinearity: require 1 and all X_{ij} to be linearly independent.
3. Normalization of error: $\mathbb{E}(u_i) = 0$.

The identification of each β_j comes from the system of equations:

$$\begin{cases} \text{Cov}(X_{i1}, u_i) = 0 \\ \text{Cov}(X_{i2}, u_i) = 0 \\ \vdots \\ \text{Cov}(X_{in}, u_i) = 0 \end{cases}$$

where we write $u_i = Y_i - \beta_0 - \sum_{j=1}^n \beta_j X_{ij}$.

The motivations for the linear regression model come from several sources. The first of which is the linear specification of non-parametric predictor, where we have $Y_i = m(X_i) + u_i$, $\mathbb{E}(u_i | X_i) = 0$ and u_i is **forecasting error**. We specify $m(X_i) = \mathbb{E}(Y_i | X_i)$. In this case, we assume $m(X_i) = \beta_0 + \beta_1 X_i$ and the conditional expectation of u_i given X_i being 0 implies the covariance between the forecasting error and X_i is 0.

The second which is that the parameters obtained are the best linear prediction of Y_i which is determined by minimizing $\mathbb{E}((Y_i - m(X_i))^2)$. For all linear m , the one that minimizes the expression is $m^*(X_i) = \beta_0 + \beta_1 X_i$, where the two parameters are in the form we specified above.

The third one is that it is a **structural causal model**, where $Y_i = g(X_i, u_i)$, and u_i this time is unobserved factor determining Y_i , only in this case, we do not know if $\text{Cov}(X_i, u_i) = 0$. In this case, X_i and Y_i determines g to be $g(x, u) = \beta_0 + \beta_1 x + u$.

2.2 Estimation

The previous section discusses identification of parameters. This section discusses that since we cannot observe the population but only a sample of the population, how we can estimate the identified parameters. Here, we give the method of **estimators**. We can estimate a parameter in many ways, therefore, it is important to determine the quality of an estimator, that is, how well it estimates the parameter of interest.

Definition 2.1. Given an estimator $\hat{\theta}$ for a parameter θ , we say that this estimator is **consistent** if

$$\forall \varepsilon > 0, n \rightarrow \infty \implies \mathbb{P}(|\hat{\theta} - \theta| > \varepsilon) \rightarrow 0$$

where n represents the size of the sample.

We want to develop an estimator $\hat{\beta}_1$ for β_1 that aligns with consistency. A straightforward way is the **sample analogue estimation**, where we replace the population form into its corresponding sample form. For sample analogue estimation to properly stand, we introduce relevant asymptotic theories.

Definition 2.2. Suppose Y_n is a sequence of random variables, and Y is a random variable, then we say

$$Y_n \rightarrow_p Y$$

Y_n converges in probability to Y , if $\forall \varepsilon > 0$,

$$\mathbb{P}(|Y_n - Y| > \varepsilon) \rightarrow 0$$

as $n \rightarrow \infty$.

We also say that Y is the probability limit of Y_i and write

$$\text{plim}_{n \rightarrow \infty} Y_n = Y$$

This also extends to random vectors, where we additionally require each element of the random vector to converge in probability to the corresponding element in the target random vector.

Proposition 2.1. Suppose $X \rightarrow_p c$ as $n \rightarrow \infty$, then for any $f \in C$ (continuous function), we have $f(X) \rightarrow_p f(c)$.

Lemma 2.1. Several useful inequalities that can assist include: $\forall 1 \leq p \leq q < \infty$

1. Jensen's Inequality: $\mathbb{E}(|X|^q) < \infty \implies \mathbb{E}(|X|^p) < \infty$.
2. Cauchy-Schwarz Inequality: $\mathbb{E}(|XY|)^2 \leq \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}$.

Theorem 2.1. The Law of Large Numbers states that, suppose $X_1, X_2, \dots, X_n$ are iid random variables, and $\mathbb{E}(|X_i|) < \infty$, then

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i \rightarrow_p \mathbb{E}(X_i)$$

Then, the sample analogue estimator of β_1 is:

$$\hat{\beta}_1 = \frac{\hat{\text{Cov}}(Y_i, X_i)}{\hat{\text{Var}}(X_i)}$$

We can obtain that:

$$\hat{\beta}_1 - \beta_1 = \frac{\hat{\text{Cov}}(X_i, u_i)}{\hat{\text{Var}}(X_i)}$$

which is the endogeneity bias.

If $\text{Cov}(X_i, u_i) = 0$, then the above will converge in probability to 0, making $\hat{\beta}_1$ consistent. We also call this the estimation error, and we can use this to conduct hypothesis testing.

2.3 Hypothesis Testing

Before we go into hypothesis testing, we introduce a few more concepts.

Definition 2.3. We say X_n converges in distribution to X (write as $X_n \rightarrow_d X$) if $\forall t \in \mathbb{R}$ such that F_X is continuous at t , we have:

$$F_{X_n}(t) \rightarrow F_X(t)$$

as $n \rightarrow \infty$.

Lemma 2.2. General Slutsky's Lemma states that suppose $X_n \rightarrow_p c$ and $Y_n \rightarrow_d Y$, then $\forall g \in C$, we have

$$g(X_n, Y_n) \rightarrow_d g(c, Y)$$

If $X_n \rightarrow_p X$, $X_n \rightarrow_d X$.

Theorem 2.2. Central Limit Theorem states that if $X_1, X_2, \dots, X_n$ are iid and $\mathbb{E}(X_i^2) < \infty$, then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mathbb{E}(X_i)) \rightarrow_d N(0, \sigma^2)$$

where

$$\sigma^2 = \text{Var}(X_i)$$

We can estimate σ^2 with its sample analogue estimator $\hat{\sigma}^2 = \hat{\text{Var}}(X_i)$.

Hypothesis testing is asking a question while expecting a “YES/NO” answer. The statement we want to discover is the **alternative hypothesis** H_1 and what we assume is the **null hypothesis** H_0 . Given a sequence of random variables $\{X_i\}_{i=1}^n$, we use it to construct a test statistic τ and a critical value c such that, if $\tau > c$, we reject H_0 .

Definition 2.4. A **Type I error** is when H_0 is true, but we have rejected H_0 .

A **Type II error** is when H_0 is false, but we failed to reject H_0 .

The test statistic is usually constructed to be:

$$\tau = \frac{\sqrt{n}(\hat{\theta} - \theta_{H_0})}{\hat{\sigma}}$$

and the critical value is $c = z_{1-\alpha}$ (this is a one-side test). Since we reject if $\tau > c$, so we care about $\mathbb{P}_{H_0}(\tau > c)$ (Type I error, assuming H_0 is true, what is the probability that $\tau > c$).

As for the estimator of β_1 , $\hat{\beta}_1$, using the above theorems, we can show that:

$$\frac{\sqrt{n}(\hat{\beta}_1 - \beta_1)}{\hat{\sigma}_\beta} \rightarrow_d N(0, 1)$$

where

$$\sigma_\beta^2 = \frac{\text{Var}(u_i(X_i - \mathbb{E}(X_i)))}{\text{Var}(X_i)^2}$$

We define **power** to be $1 - \text{Type II error} = \mathbb{P}(\text{rejecting null when null is false})$.

If we further assume **conditional homoskedasticity**, where $\mathbb{E}(u_i^2 | X_i) = \sigma^2$ for some $\sigma^2 > 0$, then the standard deviation we obtained above can be reduced to:

$$\sigma_\beta^2 = \frac{\sigma^2}{\text{Var}(X_i)}$$

where the numerator is the “noise” and the denominator is the “signal”.

The **standard error** is the square root of the asymptotic variance divided by $\sqrt{n}$. That is,

$$SE = \frac{\hat{\sigma}_\beta}{\sqrt{n}}$$

where $\hat{\sigma}_\beta^2 = \frac{\hat{\sigma}^2}{\text{Var}(X_i)}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_i \hat{u}_i^2$, with $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$. The **confidence interval** of a 2-side test is thus:

$$[\hat{\beta}_1 - z_{1-\alpha/2} \cdot SE, \hat{\beta}_1 + z_{1-\alpha/2} \cdot SE]$$

where the test statistic is $\tau = \left| \frac{\hat{\beta}_1}{SE} \right|$ and the critical value is $c = z_{1-\alpha/2}$.

This note will not go into the details, but it is good to mention the “Eicker-White Heteroskedasticity-Robust Inference” and the “Hausman-Wu Test”.

2.4 Non-linear Regression Model

We digress briefly to discuss non-linear regression models.

In general, a regression model is just $Y_i = g(X_i, u_i)$. If we define

$$m(x_1, \dots, x_k) := \mathbb{E}(Y_i | X_{i1} = x_1, \dots, X_{ik} = x_k)$$

then

Definition 2.5. The **average partial effect** is defined to be:

$$APE_{x_j} := \mathbb{E}\left(\frac{\partial}{\partial x_j} m(X_{i1}, \dots, X_{ik})\right)$$

Additionally, we can also consider it to be a “structural equation model”, which is causal. Then, we define:

Definition 2.6. The **average structure function** is defined to be:

$$ASF(x) = \int g(x, u) f_U(u) du = \mathbb{E}(g(x, U))$$

Lemma 2.3. Suppose that X and Y are independent, for any function g ,

$$\mathbb{E}(g(Y, X) \mid X = x) = \mathbb{E}(g(Y, x))$$

Then, for a structural-causal model and X_i, u_i are independent, we have:

$$APE_x = \mathbb{E}\left(\frac{\partial ASF(X_i)}{\partial x}\right)$$

For a regression model like:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + \beta_3 X_i W_i + u_i$$

where $X_i W_i$ is the interaction term, if we assume $W_i \in \{0, 1\}$, then we can interpret β_3 to be:

$$\beta_3 = \mathbb{E}\left(\frac{\partial m(X, 1)}{\partial x} - \frac{\partial m(X, 0)}{\partial x}\right)$$

2.5 Endogeneity

We return to discussing the linear regression model. Recall that we require exogeneity assumption to identify β_1 , here, we discuss different cases of endogeneity, and check if we can still identify the parameter of interest.

2.5.1 Omitted Variable Bias

The first common type of endogeneity comes from the omitted variable bias, where the real model is:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + u_i$$

but we only observe

$$Y_i = \beta_0 + \beta_1 X_i + v_i$$

where $v_i = \beta_2 W_i + u_i$. Then, the endogeneity bias is:

$$\text{plim}_{n \rightarrow \infty} \hat{\beta}_1 - \beta_1 = \frac{\text{Cov}(X_i, v_i)}{\text{Var}(X_i)} = \beta_2 \frac{\text{Cov}(X_i, W_i)}{\text{Var}(X_i)}$$

2.5.2 Measurement Error

The second common type of endogeneity bias comes from measurement errors, where the real model is

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

but we observe $\tilde{X}_i = X_i + \varepsilon_i$, where the measurement error is pure (uncorrelated with anything).

We use the model:

$$Y_i = \beta_0 + \beta_1 \tilde{X}_i + v_i$$

which gives $v_i = u_i - \beta_1 \varepsilon_i$ the component of the endogeneity bias is thus:

$$\text{Cov}(\tilde{X}_i, v_i) = -\beta_1 \text{Var}(\varepsilon_i)$$

The entire endogeneity bias is thus:

$$\frac{\text{Cov}(\tilde{X}_i, v_i)}{\text{Var}(\tilde{X}_i)} = -\beta_1 \times \frac{\text{Var}(\varepsilon_i)}{\text{Var}(X_i) + \text{Var}(\varepsilon_i)}$$

2.5.3 Simultaneity

The third common type of endogeneity bias comes from simultaneity. One case is the simultaneous causality, where we have both

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

and

$$X_i = \alpha_0 + \alpha_1 Y_i + v_i$$

we can solve for X_i and then calculate $\text{Cov}(X_i, u_i)$.

Another case is market equilibrium, where we have both

$$Q_i^D = \alpha_0 + \alpha_1 P_i^D + u_i^D$$

and

$$Q_i^S = \beta_0 + \beta_1 P_i^S + u_i^S$$

At equilibrium, we have $Q_i^D = Q_i^S = Q_i$ and $P_i^D = P_i^S = P_i$. So we have solve for P_i and it is endogenous as it contains both u_i^D and u_i^S .

2.6 Instrumental Variables

Suppose we have the same univariate linear regression model, but we do not have exogeneity. We can consider another observed variable Z_i where $\text{Cov}(Z_i, u_i) = 0$ but $\text{Cov}(X_i, Z_i) \neq 0$. Then, we can identify β_1 as the following:

$$\beta_1 = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(X_i, Z_i)}$$

The corresponding sample analogue estimator $\hat{\beta}_1$ is also a 2-series estimator, where:

$$\hat{\beta}_1 = \frac{\hat{\text{Cov}}(Y_i, Z_i)}{\hat{\text{Cov}}(X_i, Z_i)}$$

and $\sqrt{n}(\hat{\beta}_1 - \beta_1) \rightarrow_d N(0, \sigma_\beta^2)$, with

$$\sigma_\beta^2 = \frac{\text{Var}(u_i(Z_i - \mathbb{E}(Z_i)))}{\text{Cov}(X_i, Z_i)^2}$$

The reason why we use 2SLS is the following. Consider

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + u_i \\ X_i &= \pi_0 + \pi_1 Z_i + v_i \end{aligned}$$

with four assumptions: $\text{Cov}(Z_i, v_i) = 0$ (best linear prediction), $\pi_1 \neq 0$ (IV relevance), $\text{Cov}(u_i, v_i) \neq 0$ (endogeneity of X_i), and $\text{Cov}(Z_i, u_i) = 0$ (IV exogeneity). We have π_1 identified like a “normal” β_1 . Let $X_i(\pi) = \pi_0 + \pi_1 Z_i$, and regress Y_i on $X_i(\pi)$, the coefficient is thus:

$$\beta_1 = \frac{\text{Cov}(Y_i, X_i(\pi))}{\text{Var}(X_i(\pi))}$$

The **first stage** is $X_i(\hat{\pi}) = \hat{\pi}_0 + \hat{\pi}_1 Z_i$, and the **second stage** is regressing Y_i on $X_i(\hat{\pi})$.

When you have more instrumental variables than the number of endogenous regressors, we say this is **over-identification** (if the number is equal, it is **just-identified**). This is a good thing, as we have more restrictions to identify. This is particularly useful when determining whether an IV is exogenous or not.

2.6.1 IV Relevance

Say we have the model of:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + \beta_2 W_i + u_i \\ X_i &= \pi_0 + \pi_1 Z_{i1} + \pi_2 Z_{i2} + \pi_3 W_i + v_i \end{aligned}$$

We have the pair of hypotheses to be:

$$\begin{aligned} H_0 &: \pi_1 = \pi_2 = 0 \\ H_1 &: \pi_1 \neq 0 \vee \pi_2 \neq 0 \end{aligned}$$

Say we conduct an F-test and the test statistic F is way larger than the critical value and we reject H_0 . This is a good thing, as otherwise, we run into the **Weak Instrument Problem**.

This is characterized to be $\text{Cov}(X_i, Z_i) \approx 0$. The endogeneity bias is thus:

$$\begin{aligned} \hat{\beta}_1 - \beta_1 &= \frac{\hat{\text{Cov}}(u_i, Z_i) = 0}{\hat{\text{Cov}}(X_i, Z_i) \approx 0} \\ &= \frac{\sqrt{n}\hat{\text{Cov}}(u_i, Z_i) \rightarrow_d N(0, \sigma_1^2)}{\sqrt{n}\hat{\text{Cov}}(X_i, Z_i) \rightarrow_d N(0, \sigma_2^2)} \\ \hat{\beta}_1 &= \beta_1 + ??? \end{aligned}$$

Meaning we cannot construct a test statistic, because the estimator is inconsistent.

2.6.2 IV Exogeneity

If we have a just-identified model, we cannot test $\text{Cov}(Z_i, u_i)$. The estimator will always be zero, as the estimator itself is obtained by assuming $\text{Cov}(u_i, Z_i) = 0$, and here we are testing whether this

covariance is 0 or not.

However, if our model is over-identified, we can test this. Consider the model:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + \beta_2 W_i + u_i \\ X_i &= \pi_0 + \pi_1 Z_{i1} + \pi_2 Z_{i2} + \pi_3 W_i + v_i \end{aligned}$$

We have the pair of hypotheses to be:

$$\begin{aligned} H_0 : \text{Cov}(u_i, Z_{i1}) = \text{Cov}(u_i, Z_{i2}) &= 0 \\ H_1 : H_0 &\text{ not true} \end{aligned}$$

Then, we can construct:

$$\hat{u}_i = \delta_0 + \delta_1 Z_{i1} + \delta_2 Z_{i2} + \varepsilon_i$$

where $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 W_i$, and conduct a F-test of $H_0 : \delta_1 = \delta_2 = 0$.

The test statistic used here is $J = m \cdot F \rightarrow_d \chi^2_{(m-k)}$, where m is the number of IVs and k is the number of endogenous regressors.

3 Panel Data

We start by observing what panel data look like. Consider the following table:

	$t = 1$	$t = 2$	$\dots$	$t = T$
$i = 1$	Z_{11}	Z_{12}	$\dots$	Z_{1T}
$i = 2$	Z_{21}	Z_{22}	$\dots$	Z_{2T}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$i = n$	Z_{n1}	Z_{n2}	$\dots$	Z_{nT}

where $Z_{it} = (Y_{it}, X_{it})$. A column is the cross-sectional dimension, a row is the time dimension. For all $i \in \{1, \dots, n\}$, Z_{it} are independent. Note that panel data is not the same as repeated cross-section data (because it samples different individuals at each time period).

A small n and a small T “makes no sense”. A small n and a large T needs time-series analysis. A large n and a small T is a short panel, and a large n and a large T is a large panel.

3.1 Fixed Effects

Consider there is some *unobserved individual heterogeneity*, that is, there exists some time-invariant term that affects Y_{it} , but unobserved for each i . The model we use is:

$$Y_{it} = \alpha_i + \beta_1 X_{it} + u_{it}$$

The first assumption for identification is:

$$\mathbb{E}(u_{it} \mid X_{i1}, \dots, X_{iT}) = 0$$

The second assumption we make is that:

$$\sum_T \mathbb{E}((X_{it} - \bar{X}_i)^2) > 0$$

where $\bar{X}_i = \frac{1}{T} \sum_T X_{it}$ (*within group transformation*). Then, by applying this transformation, we have:

$$\begin{aligned} \bar{Y}_i &= \alpha_i + \beta_1 \bar{X}_i + \bar{u}_i \\ Y_{it} &= \alpha_i + \beta_1 X_{it} + u_{it} \end{aligned}$$

This gives that:

$$Y_{it} - \bar{Y}_i = \beta_1 (X_{it} - \bar{X}_i) + (u_{it} - \bar{u}_i)$$

equivalent to

$$Y_{it}^* = \beta_1 X_{it}^* + u_{it}^*$$

We can thus identify β_1 by

$$\mathbb{E}(Y_{it}^* X_{it}^*) = \beta_1 \mathbb{E}(X_{it}^{*2}) + \mathbb{E}(u_{it}^* X_{it}^*)$$

where $\mathbb{E}(u_{it}^* X_{it}^*) = 0$ through the assumption made above.

$$\beta_1 = \frac{\sum_T \mathbb{E}(Y_{it}^* X_{it}^*)}{\sum_T \beta_1 \mathbb{E}(X_{it}^{*2})}$$

We can directly take the expectation on both sides without multiplying X_{it}^* , but that would lead to a larger variance when estimating β_1 .

The consistent sample analogue estimator is thus:

$$\hat{\beta}_1 = \frac{\sum_T \frac{1}{n} \sum_n Y_{it}^* X_{it}^*}{\sum_T \frac{1}{n} \sum_n X_{it}^{*2}}$$

This means:

$$\hat{\beta}_1 = \beta_1 + \frac{\frac{1}{n} \sum_n \sum_T u_{it}^* X_{it}^*}{\frac{1}{n} \sum_n \sum_T (X_{it}^*)^2}$$

Then $\sqrt{n}(\hat{\beta}_1 - \beta_1) \rightarrow_d N(0, \sigma_\beta^2)$, where

$$\sigma_\beta^2 = \frac{\text{Var}(\sum_T u_{it} X_{it}^*)}{(\sum_T \mathbb{E}((X_{it}^*)^2))^2}$$

if $\mathbb{E}(\sum_T u_{it} X_{it}^*) = 0$, we also have $\text{Var}(\sum_T u_{it} X_{it}^*) = \mathbb{E}((\sum_T u_{it} X_{it}^*)^2)$.

If we also give the **conditional homoskedasticity assumption**, we have:

$$\begin{aligned} \mathbb{E}(u_{it}^2 \mid X_{i1}, \dots, X_{iT}) &= \sigma^2 \\ &= \text{Var}(u_{it} \mid X_{i1}, \dots, X_{iT}) \end{aligned}$$

as $\mathbb{E}(u_{it} \mid X_{i1}, \dots, X_{iT}) = 0$.

Additional **conditional serial uncorrelatedness** assumption gives that:

$$\mathbb{E}(u_{it} u_{is} \mid X_{i1}, \dots, X_{iT}) = 0$$

if $t \neq s$. Then, we have:

$$\sigma_\beta^2 = \frac{\sigma^2}{\sum_T \mathbb{E}((X_{it}^*)^2)}$$

Even if we have only two periods, we can still identify β_1 by subtraction between the two periods.

Another way of identifying β_1 is by **first differencing**. That is, take:

$$\begin{aligned} Y_{it} &= \alpha_i + \beta_1 X_{it} + u_{it} \\ Y_{i,t-1} &= \alpha_i + \beta_1 X_{i,t-1} + u_{i,t-1} \end{aligned}$$

and obtain:

$$\Delta Y_{it} = \beta_1 \Delta X_{it} + \Delta u_{it}$$

We then identify β_1 by

$$\mathbb{E}(\Delta Y_{it} \Delta X_{it}) = \beta_1 \mathbb{E}(\Delta X_{it}^2)$$

because $\mathbb{E}(\Delta u_{it} \Delta X_{it}) = 0$.

$$\beta_1 = \frac{\sum_{t=2}^T \mathbb{E}(\Delta Y_{it} \Delta X_{it})}{\sum_{t=2}^T \mathbb{E}(\Delta X_{it}^2)}$$

This time, the estimation variance is

$$\sigma_\beta^2 = \frac{\text{Var}(\sum_{t=2}^T \Delta u_{it} \Delta X_{it})}{(\sum_{t=2}^T \mathbb{E}(\Delta X_{it}^2))^2}$$

In a multivariate case, while fulfilling the desired assumptions, after checking relevant values to be 0, we can identify with a system of equations by taking the expected value of the product of Y_{it}^* and each $X_{it,j}^*$.

We can also introduce instrumental variables where we require:

1. $\mathbb{E}(u_{it} \mid Z_{i1}, \dots, Z_{iT}) = 0$, and
2. $\sum_{t=1}^T \mathbb{E}(X_{it}^* Z_{it}^*) \neq 0$.

We identify β_1 by:

$$\beta_1 = \frac{\sum_T \mathbb{E}(Y_{it}^* Z_{it}^*)}{\sum_T \mathbb{E}(X_{it}^* Z_{it}^*)}$$

4 Causal Inference

We analyze causal models using potential outcomes, that is analyzing the causality between Y_i and $D_i \in \{0, 1\}$. We call $Y_i(1)$ and $Y_i(0)$ to be the potential outcomes.

The **treatment effect** is thus (causal effect of D_i): $Y_i(1) - Y_i(0)$. This leads to the **average treatment effect** being

$$ATE = \mathbb{E}(Y_i(1) - Y_i(0))$$

We observe the following:

$$Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i) = D_i(Y_i(1) - Y_i(0)) + Y_i(0)$$

Then, if we introduce D_i to be treatment status, and X_i to be some covariate, under the **randomized control trial assumption**, that is $(Y_i(1), Y_i(0), X_i)$ are all independent of D_i , we have:

$$ATE = \frac{\text{Cov}(Y_i, D_i)}{\text{Var}(D_i)} = \mathbb{E}(Y_i \mid D_i = 1) - \mathbb{E}(Y_i \mid D_i = 0) = \frac{\mathbb{E}(Y_i D_i)}{\mathbb{P}(D_i = 1)} - \frac{\mathbb{E}(Y_i(1 - D_i))}{\mathbb{P}(D_i = 0)}$$

If we wish to test things about ATE, notice that we can take $W_i = \frac{Y_i D_i}{p} - \frac{Y_i(1 - D_i)}{1 - p}$, because we often know what p , and the test statistic would be

$$\tau = \frac{\sqrt{n}(\bar{W} - ATE_{H_0})}{\sqrt{\hat{\text{Var}}(W_i)}}$$

4.1 Selection on Observation

Here, we no longer assume that the covariate X_i is independent of D_i . However, we still need to make two assumptions to identify relevant parameters.

The first one is **unconfoundedness** assumption, where $(Y_i(1), Y_i(0))$ are independent of D_i given X_i .

The second one is **overlap condition**, where

$$\exists \varepsilon > 0, \varepsilon < \mathbb{P}(D_i = 1 \mid X_i = x) < 1 - \varepsilon$$

where we call $\mathbb{P}(D_i = 1 \mid X_i = x)$ the **propensity score**.

We introduce relevant properties of **conditional independence**.

Lemma 4.1. Suppose that V and W are independent given X , then for any function ϕ :

$$\mathbb{E}(\phi(V) \mid W = w, X = x) = \mathbb{E}(\phi(V) \mid X = x)$$

If Y is independent of (X, W) , then Y is independent of X given W .

If Y and X are independent given W , then $f(Y)$ and $g(X)$ are independent given W .

The overlap condition implies that the $\text{supp}(X_i \mid D_i = 1) = \text{supp}(X_i \mid D_i = 0)$.

ATE is thus identified to be:

$$ATE = \frac{\mathbb{E}(Y_i D_i)}{\mathbb{P}(D_i = 1 \mid X_i)} - \frac{\mathbb{E}(Y_i(1 - D_i))}{\mathbb{P}(D_i = 0 \mid X_i)}$$

This can take a sample analogue estimator given that defining $p(x) = \mathbb{P}(D_i = 1 \mid X_i = x)$ and $\hat{p}(x) \rightarrow_p p(x)$. This is an “inverse propensity score weighted estimator”.

4.2 Difference-in-differences

Suppose we have $Y_{it}(d)$, where $t = 0, 1$ and $d = 0, 1$. We observe:

$$Y_{it} = D_i Y_{it}(1) + (1 - D_i) Y_{it}(0) = D_i(Y_{it}(1) - Y_{it}(0)) + Y_{it}(0)$$

and we define *average treatment effect on the treated* to be:

$$ATT = \mathbb{E}(Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1)$$

$t = 0$ means before treatment is applied, and $t = 1$ means after treatment is applied.

We make two assumptions:

1. No anticipation: $Y_{i0}(1) = Y_{i0}(0)$.
2. Parallel trend assumption:

$$\mathbb{E}(Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1) = \mathbb{E}(Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0)$$

We want to identify ATT. Note that by “no anticipation”, we have $Y_{i0} = Y_{i0}(0)$. Then, by $Y_{i1} - Y_{i0}$ and take the expected value given $D_i = 0, 1$, we have:

$$ATT = \mathbb{E}(Y_{i1} - Y_{i0} \mid D_i = 1) - \mathbb{E}(Y_{i1} - Y_{i0} \mid D_i = 0)$$

Additionally, we have:

$$\begin{aligned} ATT &= \mathbb{E}(Y_{i1} - Y_{i0} \mid D_i = 1) - \mathbb{E}(Y_{i1} - Y_{i0} \mid D_i = 0) \\ &= \frac{\mathbb{E}(\Delta Y_i D_i)}{\mathbb{E}(D_i)} - \frac{\mathbb{E}(\Delta Y_i (1 - D_i))}{\mathbb{E}(1 - D_i)} \end{aligned}$$

and then, we can provide a sample analogue estimator:

$$\hat{ATT} = \frac{\hat{\text{Cov}}(\Delta Y_i, D_i)}{\hat{\text{Var}}(D_i)}$$

4.2.1 Two-way Fixed Effects

Say we have:

$$Y_{it} = \alpha_i + \gamma_t + \beta \cdot 1\{t = 1\} \cdot D_i + u_{it}$$

with $\mathbb{E}(u_{it} \mid \alpha_i, D_i) = 0$. β can be identified in a similar way where:

$$\beta = \mathbb{E}(\Delta Y_i \mid D_i = 1) - \mathbb{E}(\Delta Y_i \mid D_i = 0)$$

As more multi-period two-way fixed effects, which is not discussed in this scope of the course, are discussed in many papers.

4.2.2 Involvement of Covariate

To do DID with covariance, we still have no anticipation assumption, but we update to *conditional parallel trend assumption* where:

$$\mathbb{E}(Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1, X_i = x) = \mathbb{E}(Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0, X_i = x)$$

The conditional ATT is thus:

$$CATT = \mathbb{E}(Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1, X_i = x)$$

In this case, we call:

$$CATT(x) = \mathbb{E}(\Delta Y_i \mid D_i = 1, X_i = x) - \mathbb{E}(\Delta Y_i \mid D_i = 0, X_i = x)$$

Then, ATT is:

$$\mathbb{E}(CATT(X_i) \mid D_i = 1) = \mathbb{E}(\Delta Y_i \mid D_i = 1) - \mathbb{E}(\mathbb{E}(\Delta Y_i \mid D_i = 0, X_i) \mid D_i = 1)$$

4.2.3 Estimating ATT: Linear Regression with Interaction Term

Call

$$(Y_{10}, Y_{20}, \dots, Y_{n0}, Y_{11}, \dots, Y_{n1}) = (\tilde{Y}_1, \dots, \tilde{Y}_{2n})$$

Define T_j to be 0 if $\tilde{Y}_j$ comes from period 0 and to be 1 if $\tilde{Y}_j$ comes from period 1. Define $\tilde{D}_j$ to be 0 if $\tilde{Y}_j$ comes from the control group and to be 1 otherwise.

We write $\gamma_{dt} = \mathbb{E}(\tilde{Y}_j \mid \tilde{D}_j = d, T_j = t)$, and

$$\mathbb{E}(\tilde{Y}_j \mid \tilde{D}_j, T_j) = \beta_0 + \beta_1 \tilde{D}_j + \beta_2 T_j + \beta_3 \tilde{D}_j T_j$$

Then

$$\beta_3 = (\mathbb{E}(\tilde{Y}_j \mid \tilde{D}_j = 1, T_j = 1) - \mathbb{E}(\tilde{Y}_j \mid \tilde{D}_j = 1, T_j = 0)) - (\mathbb{E}(\tilde{Y}_j \mid \tilde{D}_j = 0, T_j = 1) - \mathbb{E}(\tilde{Y}_j \mid \tilde{D}_j = 0, T_j = 0))$$

This gives $\beta_3 = \gamma_{11} - \gamma_{10} - (\gamma_{01} - \gamma_{00})$.

Then $\beta_3 = ATT$ if we write $\tilde{Y}_j = \mathbb{E}(\tilde{Y}_j \mid \tilde{D}_j, T_j) + u_j$